IA Projects

General Guidelines:

- Teams composed from 5-6 students who are registered for the course.
- Each team's Application should satisfy its software requirements.
- Backend should be in (.Net Core Web API) and the Frontend should be in React.js.
- Every team member will be questioned regarding any frontend or backend related matter
- Any Code written without understanding will allow student to get Zero in project discussion.

1- VendorHub

An e-commerce platform that allows vendors to sell products online while providing analytics and management tools for their stores.

Actors (3 types of users):

- Admin
- Vendor
- Customer

Requirements:

- Different actors can log in/log out. A customer can view products without logging in but cannot purchase products unless they log in.
- Admin can manage (accept or reject) newly registered accounts (Vendors).
- Admin can accept or reject product posts created by vendors before they become visible to customers.
- Admin can enable or disable specific permissions for vendor roles or individual vendors.
- Customer can browse all approved products and search based on category, price, or product name.
- Product post includes: Vendor Name, Title, Description, Price, Category, Images, Available Units, and Number of Viewers.
- Customer can purchase one or more units of a product if it is available in stock.
- Customer can add products to a favorites list.
- Customer can submit ratings and reviews for purchased products.
- Vendor can manage products (create, list, update, delete) (CRUD).
- Vendor can update product stock and pricing.
- Vendor can view sales history and product performance statistics.
- Vendor receives a notification when a customer purchases one of their products.

Notes:

- **You must use real-time sockets (e.g., purchase notifications).**
- **DB design (Schema) is required**

2- BookCircle

A community-driven book sharing platform where users can lend, borrow, and discuss books.

Actors (3 types of users):

- Admin
- Book Owner
- Reader

Requirements:

- Different actors can log in/log out. A reader can view available books without logging in but cannot borrow them unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Book Owners).
- Admin can accept or reject book posts before they appear in the system.
- Reader can browse available books and search based on genre, language, or borrow price.
- Book post includes: Book Owner Name, Title, Genre, ISBN, Language, Publication Date, Borrow Price, Borrow Status (Available / Borrowed), Availability Dates, and Cover Image.
- Reader can send a borrow request to the book owner.
- Book Owner can accept or reject borrow requests.
- When accepted, the book status changes to Borrowed.
- Reader can like or dislike books.
- Readers can write comments on book posts, with a single level of replies.
- Users can create reading lists to save books they are interested in.
- Book Owner can manage book posts (create, list, update, delete) (CRUD).
- Book Owner cannot delete a book post if it is currently borrowed.

Notes:

- **You must use real-time sockets (e.g., notifications for borrow requests and comments).**
- **DB design (Schema) is required**

3- TaskFlow

A project management system that helps teams organize tasks and track project progress.

Actors (3 types of users):

- Admin
- Project Manager
- Team Member

Requirements:

- Different actors can log in/log out.
- Admin can manage (accept or reject) newly registered accounts (Project Managers).
- Project Manager can create and manage projects.
- Each project can contain multiple tasks.
- Task includes: Title, Description, Assigned Member, Priority, Due Date, and Status.
- Tasks can move between workflow stages such as To Do, In Progress, and Done.
- Project Manager can assign tasks to team members.
- Team members can update task progress and add comments.
- Team members can attach files to tasks.
- Project Manager can view project progress statistics and completed tasks.
- Team members receive notifications when tasks are assigned or updated.

Notes:

- **You must use real-time sockets as an example.**
- **DB design (Schema) is required**

4- PetAdopt

A platform that connects animal shelters and pet owners with people who want to adopt pets.

Actors (3 types of users):

- Admin
- Shelter / Pet Owner
- Adopter

Requirements:

- Different actors can log in/log out. An adopter can browse pets without logging in but cannot apply for adoption unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Shelters or Pet Owners).
- Admin can approve or reject pet posts before they appear publicly.
- Adopters can browse pets and search based on animal type, breed, age, or location.
- Pet post includes: Owner Name, Pet Name, Age, Breed, Gender, Health Status, Description, Location, and Images.
- Adopters can submit adoption requests.
- Pet owners can accept or reject adoption requests.
- Once accepted, the pet status changes to Adopted.
- Adopters can save pets to a favorites list.
- Users who successfully adopt pets can submit reviews and feedback for the shelter or owner.
- Shelter / Pet Owner can manage pet posts (create, list, update, delete) (CRUD).

Notes:

- **You must use real-time sockets as an example.**
- **DB design (Schema) is required**

5- SmartRent

A property rental platform that connects landlords and tenants and allows tenants to schedule property visits before renting.

Actors (3 types of users):

- Admin
- Landlord
- Tenant

Requirements:

- Different actors can log in/log out. A tenant can view properties without logging in but can’t interact with them unless they log in.
- Admin can manage (accept or reject) newly registered accounts (Landlords).
- Admin can accept or reject property posts created by landlords before they become visible.
- Tenant can browse all approved property listings and search based on location, price, or property type.
- Property post includes: Landlord Name, Title, Description, Price, Location, Images, Property Type, Amenities (Parking, Elevator, Furnished), and Rental Status.
- Tenant can schedule a property visit appointment with the landlord.
- Landlord can accept or reject visit requests.
- Tenant can submit a rental application including required documents.
- Landlord can review rental applications and accept or reject them.
- Once an application is accepted, the property status changes to Rented.
- Tenant can add properties to a favorites list.
- Tenant can rate and review the property after completing the rental period.
- Landlord can manage property listings (create, list, update, delete) (CRUD).

Notes:

- **You must use real-time sockets (e.g., for visit requests and notifications).**
- **DB design (Schema) is required**

6- DriveShare

A peer-to-peer car rental platform where car owners can rent their vehicles to verified users using a booking calendar.

Actors (3 types of users):

- Admin
- Car Owner
- Renter

Requirements:

- Different actors can log in/log out. A renter can view available cars without logging in but cannot rent them unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Car Owners).
- Admin can accept or reject car posts created by owners before they appear in listings.
- Renter can browse cars and search based on car type, brand, location, or price.
- Car post includes: Owner Name, Title, Description, Car Type, Brand, Model, Year, Transmission (Automatic / Manual), Location, Rental Price, Availability Calendar, and Rental Status.
- Renter can request a rental for specific dates.
- Car Owner can accept or reject rental requests.
- Renter must submit driver license verification before booking.
- Once a request is accepted, the car is marked as Rented for the selected dates.
- Renters can rate the car and leave feedback after the rental is completed.
- Car Owner can manage car posts (create, list, update, delete) (CRUD).
- Car Owner cannot delete a car post if the car is currently rented.

Notes:

- **You must use real-time sockets (e.g., rental request notifications).**
- **DB design (Schema) is required**

7- EventHub

A modern event planning and ticket management platform that helps organizers manage events and allows participants to discover and attend them.

Actors (3 types of users):

- Admin
- Event Organizer
- Participant

Requirements:

- Different actors can log in/log out. A participant can view events without logging in but can’t interact with them unless they log in.
- Admin can manage (accept or reject) newly registered accounts (Event Organizers).
- Admin can accept or reject event posts created by event organizers. Accepted events become visible to participants.
- Participant can view all available events and search for events based on criteria such as location, category, or date.
- Event post includes: Organizer name, Title, Description, Venue, Category, Event Date, Ticket Price, Total Tickets, Available Tickets, and Event Image.
- Participant can purchase event tickets if tickets are available.
- Each purchased ticket generates a unique digital ticket with a QR code.
- Participant can save events to a favorites list (watchlist).
- Participant can rate and review events after attending them.
- Event Organizer can manage events (create, list, update, delete) (CRUD).
- Event Organizer can upload attachments such as event materials or schedules.
- Event Organizer can view event analytics such as number of tickets sold and total revenue.
- Participants receive notifications when updates are posted by organizers.

Notes:

- **You must use real-time sockets (e.g., for ticket updates and notifications).**
- **DB design (Schema) is required**

8- SkillBridge

A freelancing platform where service providers offer small services and clients can hire them.

Actors (3 types of users):

- Admin
- Service Provider
- Client

Requirements:

- Different actors can log in/log out. A client can view available services without logging in but cannot request services unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Service Providers).
- Admin can accept or reject service posts created by providers before they appear publicly.
- Clients can browse services and search based on category, price, or rating.
- Service post includes: Provider Name, Service Title, Description, Category, Price, Delivery Time, and Portfolio Images.
- Client can send service requests or proposals.
- Service Provider can accept or reject service requests.
- After completing the service, clients can submit ratings and reviews.
- Service Providers can manage services (create, list, update, delete) (CRUD).
- Clients and providers can communicate through a private messaging system

Notes:

- **You must use real-time sockets as an example.**
- **DB design (Schema) is required**

9- StudyGroup

A platform for students to create and join study groups for collaborative learning.

Actors (3 types of users):

- Admin
- Group Creator
- Student

Requirements:

- Different actors can log in/log out. A student can browse study groups without logging in but cannot join them unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Group Creators).
- Admin can approve or reject study groups created by group creators.
- Students can browse study groups and search based on subject, location, or meeting time.
- Study group includes: Creator Name, Subject, Description, Maximum Members, Meeting Type (Online / Offline), Meeting Schedule, and Group Materials.
- Students can request to join a study group.
- Group Creator can accept or reject join requests.
- Group members can share study materials and files.
- Group members can participate in group discussions and comments.
- Group Creator can manage groups (create, list, update, delete) (CRUD).

Notes:

- **You must use real-time sockets as an example.**
- **DB design (Schema) is required**

10- ArtAuction

An online art auction platform where artists can showcase their artworks and buyers can place bids in real time.

Actors (3 types of users):

- Admin
- Artist
- Buyer

Requirements:

- Different actors can log in/log out. A buyer can view artworks without logging in but cannot place bids unless logged in.
- Admin can manage (accept or reject) newly registered accounts (Artists).
- Admin can accept or reject artwork posts before they become visible to buyers.
- Buyer can browse artworks and filter them based on artist name, category, or tags.
- Artwork post includes: Artist Name, Title, Description, Initial Price, Buy Now Price (optional), Auction Start Time, Auction End Time, Category, Tags, and Artwork Image.
- Buyers can place bids during the auction period.
- Each new bid must be at least $10 higher than the previous bid.
- Buyers can add artworks to a watchlist to track auctions.
- Any visitor can view the bid history including bidder name, price, and timestamp.
- Artist can manage artworks (create, list, update, delete) (CRUD).
- Artist can extend the auction end time if necessary.
- After the auction ends, the system automatically determines the winning bidder.
- The winner receives a notification with the purchase details

Notes:

- **You must use real-time sockets as an example.**
- **DB design (Schema) is required**